

Homework I

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1 Question I

Given the Lagrangian of a unit mass harmonic oscillator of frequency ω

$$L = \frac{1}{2}\dot{q}^2 - \frac{1}{2}q^2\omega^2 \quad (1.1)$$

1.1 Part (a)

If we split $q(t) = q_{\text{cl}}(t) + \tilde{q}$, we have

$$L' = \frac{1}{2}\dot{q}_{\text{cl}}^2 + \frac{1}{2}\dot{\tilde{q}}^2 + \dot{q}_{\text{cl}}\dot{\tilde{q}} - \frac{1}{2}[q_{\text{cl}}^2(t) + 2q_{\text{cl}}\tilde{q} + \tilde{q}^2]\omega^2 \quad (1.2)$$

Now we can consider the Euler-Lagrange equations

$$\frac{d}{dt}\frac{dL}{d\dot{q}} - \frac{dL}{dq} = 0 \quad (1.3)$$

Applying to our Lagrangian gives:

$$\begin{aligned} \frac{d}{dt}[\dot{q}] - q\omega^2 &= 0 \\ \ddot{q} + q\omega^2 &= 0. \end{aligned}$$

We have a second order ODE, so we can solve it using the standard exponential. Let $q = e^{\lambda t}$, thus $\ddot{q} = \lambda^2 e^{\lambda t}$

$$\begin{aligned} (\lambda^2 + \omega^2)e^{\lambda t} &= 0 \\ \lambda^2 &= -\omega^2 \implies \lambda = \pm i\omega. \end{aligned}$$

So the most general solution is

$$q(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}, \quad C_1, C_2 \in \mathbb{C}.$$

We have that $e^{\pm i\omega t} = \cos \omega t \pm i \sin \omega t$, so

$$\begin{aligned} q(t) &= C_1 (\cos \omega t + i \sin \omega t) + C_2 (\cos \omega t - i \sin \omega t) \\ &= (C_1 + C_2) \cos \omega t + i(C_1 - C_2) \sin \omega t \\ q &= A \cos \omega t + B \sin \omega t. \end{aligned}$$

We can define $A \equiv C_1 + C_2$ and $B \equiv i(C_1 - C_2)$.

$$\ddot{q} = -\omega^2 (A \cos \omega t + B \sin \omega t) \quad (1.4)$$

Since $q(t)$ is a physical coordinate it must be real, thus, we have the extra constraint that $C_1^* = C_2$ which leaves $A = 2\Re C_1$ and $B = -2\Im C_1$, both of which are purely real. Verifying the solution:

$$[-\omega^2 (A \cos \omega t + B \sin \omega t)] + [A \cos \omega t + B \sin \omega t] \omega^2 = 0.$$

We can determine A and B in terms of q_a , q_b , ω and T . Since $q(0) = q_{\text{cl}}(0) = q_a$ we have that $A = q_a$. At $t = T$, $q(T) = q_b = A \cos \omega T + B \sin \omega T$

$$\begin{aligned} q_b &= q_a \cos \omega T + B \sin \omega T \\ B &= \frac{q_b - q_a \cos \omega T}{\sin \omega T}. \end{aligned}$$

Explicitly,

$$q_{\text{cl}}(t) = q_a \cos \omega t + \left[\frac{q_b - q_a \cos \omega T}{\sin \omega T} \right] \sin \omega t$$

1.2 Part (b)

We want to show

$$S[q] = S[q_{\text{cl}} + \tilde{q}] = S[q_{\text{cl}}] + S[\tilde{q}] \quad (1.5)$$

As we had in (1.2)

$$L' = \frac{1}{2} \dot{q}_{\text{cl}}^2 + \frac{1}{2} \dot{\tilde{q}}^2 + \dot{q}_{\text{cl}} \dot{\tilde{q}} - \frac{1}{2} [q_{\text{cl}}^2(t) + 2q_{\text{cl}}\tilde{q} + \tilde{q}^2] \omega^2 \quad (1.6)$$

$$S[q] = \int_0^T dt L' = S[q_{\text{cl}} + \tilde{q}] = S[q_{\text{cl}}] + S[\tilde{q}] + \int_0^T dt [\dot{q}_{\text{cl}} \dot{\tilde{q}} - q_{\text{cl}} \tilde{q} \omega^2] \quad (1.7)$$

Thus we need to show the integral vanishes. Let $u = \dot{q}_{\text{cl}}$ and $v = \tilde{q}$

$$\int_0^T dt [\dot{q}_{\text{cl}} \dot{\tilde{q}}] = [\dot{q}_{\text{cl}} \tilde{q}] \Big|_0^T - \int_0^T \tilde{q} \ddot{q}_{\text{cl}} dt \quad (1.8)$$

But we have that $\tilde{q}(0) = 0$ and $\tilde{q}(T) = 0$. So the first term in (1.8) vanishes. We have the integral

$$- \int_0^T dt [\tilde{q} \ddot{q}_{\text{cl}} + q_{\text{cl}} \tilde{q} \omega^2] = - \int_0^T dt \tilde{q} [\ddot{q}_{\text{cl}} + q_{\text{cl}} \omega^2] \quad (1.9)$$

which vanishes by the equation of motion.

1.3 Part (c)

We have

$$S[q_{\text{cl}}] = \int_0^T dt \left[\frac{1}{2} \dot{q}_{\text{cl}}^2 - \frac{1}{2} q_{\text{cl}}^2 \omega^2 \right] \quad (1.10)$$

where $q_{\text{cl}}(t) = q_a \cos \omega t + \left[\frac{q_b - q_a \cos \omega T}{\sin \omega T} \right] \sin \omega t$, but I will revert back to A and B for brevity.

$$\begin{aligned} S[q_{\text{cl}}] &= \int_0^T dt \left[\frac{\omega^2}{2} [-A \sin \omega t + B \cos \omega t]^2 - \frac{1}{2} [A \cos \omega t + B \sin \omega t]^2 \omega^2 \right] \\ &= \int_0^T dt \frac{\omega^2}{2} [A^2 \sin^2 \omega t + B^2 \cos^2 \omega t - 2AB \sin \omega t \cos \omega t - A^2 \cos^2 \omega t - B^2 \sin^2 \omega t - 2AB \sin \omega t \cos \omega t] \\ &= \int_0^T dt \frac{\omega^2}{2} [(A^2 - B^2) [\sin^2 \omega t - \cos^2 \omega t] - 4AB \sin \omega t \cos \omega t]. \end{aligned}$$

Using $\cos^2 \omega t = \frac{1}{2}(1 + \cos 2\omega t)$ and $\sin^2 \omega t = \frac{1}{2}(1 - \cos 2\omega t)$ and $2 \sin \omega t \cos \omega t = \sin(2\omega t)$.

$$\begin{aligned} S[q_{cl}] &= \int_0^T dt \frac{\omega^2}{2} [-[A^2 - B^2] \cos 2\omega t - 2AB \sin 2\omega t] \\ &= \frac{\omega^2}{2} \left[-[A^2 - B^2] \frac{1}{2\omega} \sin 2\omega t + \frac{AB}{\omega} \cos 2\omega t \right] \Big|_0^T \\ &= \frac{\omega}{4} [-[A^2 - B^2] \sin 2\omega T + 2AB (\cos 2\omega T - 1)]. \end{aligned}$$

Substituting A and B :

$$S[q_{cl}] = \frac{\omega}{4} \left[- \left[q_a^2 - \left(\frac{q_b - q_a \cos \omega T}{\sin \omega T} \right)^2 \right] \sin 2\omega T + 2q_a \left(\frac{q_b - q_a \cos \omega T}{\sin \omega T} \right) (\cos 2\omega T - 1) \right].$$

Simplifying

$$\begin{aligned} \frac{4}{\omega} S[q_{cl}] &= -q_a^2 \sin 2\omega T + (q_b^2 - 2q_a q_b \cos \omega T + q_a^2 \cos^2 \omega T) \frac{\sin 2\omega T}{\sin^2 \omega T} \\ &\quad + \frac{2q_a}{\sin \omega T} (q_b \cos 2\omega T - q_a \cos \omega T \cos 2\omega T - q_b + q_a \cos \omega T) \\ \frac{4}{\omega} S[q_{cl}] &= -q_a^2 \sin 2\omega T + (q_b^2 - 2q_a q_b \cos \omega T + q_a^2 \cos^2 \omega T) \frac{2 \sin \omega T \cos \omega T}{\sin^2 \omega T} \\ &\quad + \frac{2q_a}{\sin \omega T} (q_b (1 - 2 \sin^2 \omega T) - q_a \cos \omega T \cos 2\omega T - q_b + q_a \cos \omega T). \end{aligned}$$

where we used the above identities to obtain the last line,

$$\begin{aligned} \frac{4}{\omega} S[q_{cl}] &= -2q_a^2 \sin \omega T \cos \omega T + (q_b^2 - 2q_a q_b \cos \omega T + q_a^2 \cos^2 \omega T) \frac{2 \cos \omega T}{\sin \omega T} \\ &\quad + \frac{2q_a}{\sin \omega T} (q_b - 2q_b \sin^2 \omega T - q_a \cos \omega T (1 - 2 \sin^2 \omega T) - q_b + q_a \cos \omega T) \\ \frac{4}{\omega} S[q_{cl}] &= -2q_a^2 \sin \omega T \cos \omega T + (q_b^2 - 2q_a q_b \cos \omega T + q_a^2 \cos^2 \omega T) \frac{2 \cos \omega T}{\sin \omega T} \\ &\quad + \frac{2q_a}{\sin \omega T} (-2q_b \sin^2 \omega T + 2q_a \cos \omega T \sin^2 \omega T). \end{aligned}$$

Multiplying by $\frac{\sin \omega T}{2}$

$$\begin{aligned} \frac{4 \sin \omega T}{2\omega} S[q_{cl}] &= -q_a^2 \sin^2 \omega T \cos \omega T + \cos \omega T (q_b^2 - 2q_a q_b \cos \omega T + q_a^2 \cos^2 \omega T) \\ &\quad - 2q_a q_b \sin^2 \omega T + 2q_a^2 \cos \omega T \sin^2 \omega T \\ \frac{2 \sin \omega T}{\omega} S[q_{cl}] &= -q_a^2 \sin^2 \omega T \cos \omega T + \cos \omega T q_b^2 - 2q_a q_b \cos^2 \omega T + q_a^2 \cos^3 \omega T \\ &\quad - 2q_a q_b \sin^2 \omega T + 2q_a^2 \cos \omega T \sin^2 \omega T \\ \frac{2 \sin \omega T}{\omega} S[q_{cl}] &= \cos \omega T q_b^2 - 2q_a q_b + q_a^2 \cos^3 \omega T + q_a^2 \cos \omega T \sin^2 \omega T \\ \frac{2 \sin \omega T}{\omega} S[q_{cl}] &= \cos \omega T q_b^2 - 2q_a q_b + q_a^2 \cos \omega T. \end{aligned}$$

where we used $\cos^2 A + \sin^2 A = 1$. Finally we get

$$S[q_{cl}] = \frac{\omega}{2 \sin \omega T} [(q_a^2 + q_b^2) \cos \omega T - 2q_a q_b] \quad (1.11)$$

In the limit of a free particle, $\omega \rightarrow 0$:

We can use the fact that for $x \ll 1$, $\cos x = 1 - \frac{1}{2}x^2$ and $\sin x = x - \frac{1}{6}x^3 + \dots$

$$S[q_{\text{cl}}] = \frac{\omega}{2\omega T \left[1 - \frac{1}{6}\omega^2 T^2 + \dots\right]} \left[(q_a^2 + q_b^2) \left(1 - \frac{1}{2}\omega^2 T^2 + \dots\right) - 2q_a q_b \right] \quad (1.12)$$

We can truncate the prefactor

$$\frac{\omega}{2\omega T \left[1 - \frac{1}{6}\omega^2 T^2 + \dots\right]} \sim \frac{1}{2T}.$$

Likewise, for the second factor

$$1 - \frac{1}{2}\omega^2 T^2 + \dots \sim 1 - \frac{1}{2}\omega^2 T^2.$$

$$\begin{aligned} S(q_{\text{cl}}) &= \frac{1}{2T} \left[(q_a^2 + q_b^2) \left(1 - \frac{1}{2}\omega^2 T^2\right) - 2q_a q_b \right] \\ S(q_{\text{cl}}) &= \frac{1}{2T} \left[(q_a^2 + q_b^2) - \frac{q_a^2 + q_b^2}{2}\omega^2 T^2 - 2q_a q_b \right] \\ S(q_{\text{cl}}) &= \frac{1}{2T} (q_a - q_b)^2 - \frac{q_a^2 + q_b^2}{4}\omega^2 T \\ S(q_{\text{cl}}) &= \frac{1}{2T} (q_a - q_b)^2 + \mathcal{O}(\omega^2) \end{aligned}$$

1.4 Part (d)

We have the definition of the Fourier expansion of an arbitrary piecewise smooth periodic function with period $2T$

$$f(t) = a_0 + \sum_{n \geq 1} a_n \sin\left(\frac{n\pi t}{T}\right) + b_n \cos\left(\frac{n\pi t}{T}\right) \quad (1.13)$$

We can let $f(t) = \tilde{q}(t)$ where we are given the initial conditions: $\tilde{q}(0) = 0$, $\tilde{q}(T) = 0$ and $\tilde{q}(t)$ is only defined on the interval $[0, T]$ and since the domain doesn't extend past one period, it cannot have periodic structure. We cannot apply the Fourier expansion to $\tilde{q}$ directly, we need to extend its domain.

We can extend the domain by considering $[-T, 0]$ and demanding that $\tilde{q}(-t) = -\tilde{q}(t)$. Thus it is periodic with period $2T$ and has the domain $[-T, T]$. We can verify it obeys the boundary conditions $\tilde{q}(0) = -\tilde{q}(0) = 0$ and $\tilde{q}(-T) = -\tilde{q}(T) = 0$. The boundary condition of $\tilde{q}(0) = 0$ implies $a_0 = 0$. Since we have noticed it is odd we demand $b_n = 0$ since cosine is an even function we can't have any even contribution.

Thus we have

$$\tilde{q}(t) = \sum_{n \geq 1} a_n \sin\left(\frac{n\pi t}{T}\right) \quad (1.14)$$

If we take the time derivative

$$\dot{q}(t) = \sum_{n \geq 1} a_n \frac{n\pi}{T} \cos\left(\frac{n\pi t}{T}\right) \quad (1.15)$$

Now we can consider the action

$$\begin{aligned} e^{iS[\tilde{q}]} &= \exp\left[\frac{i}{2} \left(\int dt [\dot{\tilde{q}}^2 - \tilde{q}^2 \omega^2] \right)\right] \\ \dot{\tilde{q}}^2 &= \sum_{n,m=1}^{\infty} a_n a_m \left(\frac{nm\pi^2}{T^2} \right) \cos\left(\frac{n\pi t}{T}\right) \cos\left(\frac{m\pi t}{T}\right) \\ \tilde{q}^2 &= \sum_{n,m=1}^{\infty} a_n a_m \sin\left(\frac{n\pi t}{T}\right) \sin\left(\frac{m\pi t}{T}\right). \end{aligned}$$

For $n = m$ the exponent takes the form

$$\frac{i}{2} \sum_{n=1}^{\infty} a_n^2 \int dt \left[\frac{n^2 \pi^2}{T^2} \cos^2\left(\frac{n\pi t}{T}\right) - \omega^2 \sin^2\left(\frac{n\pi t}{T}\right) \right].$$

Using

$$\begin{aligned} \cos^2 A &= \frac{1}{2} (1 + \cos 2A) \\ \sin^2 A &= \frac{1}{2} (1 - \cos 2A). \end{aligned}$$

$$\begin{aligned} &\frac{i}{4} \sum_{n=1}^{\infty} a_n^2 \int dt \left[\frac{n^2 \pi^2}{T^2} \left[1 + \cos \frac{2\pi n t}{T} \right] - \omega^2 \left(1 - \cos \frac{2\pi n t}{T} \right) \right] \\ &\frac{i}{4} \sum_{n=1}^{\infty} a_n^2 \left[\frac{n^2 \pi^2}{T^2} \left[t + \frac{T}{2\pi n} \sin \frac{2\pi n t}{T} \right] - \omega^2 \left(t - \frac{T}{2\pi n} \sin \frac{2\pi n t}{T} \right) \right] \Big|_{t=0}^{t=T} \\ &\frac{i}{4} \sum_{n=1}^{\infty} a_n^2 \left[\frac{n^2 \pi^2}{T^2} \left[T + \frac{T}{2\pi n} \sin 2\pi n \right] - \omega^2 \left(T - \frac{T}{2\pi n} \sin 2\pi n \right) \right] \\ &\frac{i}{4} \sum_{n=1}^{\infty} a_n^2 \left[\frac{n^2 \pi^2}{T} - \omega^2 T \right] \end{aligned}$$

where we used that sine vanishes at integer multiples of π .

Since the exponential to a sum is a product of exponentials we have

$$e^{iS[\tilde{q}]} = \prod_{n=1}^{\infty} \exp \left[-a_n^2 \left(-\frac{in^2 \pi^2}{4T} \left(1 - \frac{\omega^2 T^2}{n^2 \pi^2} \right) \right) \right].$$

If we define $l_n := \frac{-in^2 \pi^2}{4T} \left(1 - \frac{\omega^2 T^2}{n^2 \pi^2} \right)$. We have

$$e^{iS[\tilde{q}]} = \prod_{n=1}^{\infty} \exp [-a_n^2 l_n] \quad (1.16)$$

as required.

1.5 Part (e)

From part (c) we had

$$\begin{aligned}
\left\langle q_b, T \middle| q_a, 0 \right\rangle &= \int [Dq] e^{iS[q]} = e^{iS[q_{cl}]} \int [D\tilde{q}] e^{iS[\tilde{q}]} \\
\left\langle q_b, T \middle| q_a, 0 \right\rangle &= e^{iS[q_{cl}]} \int [D\tilde{q}] e^{iS[\tilde{q}]} \\
&= e^{iS[q_{cl}]} \int \left[C \prod_{n=1}^{\infty} da_n \right] e^{-l_n a_n^2} \\
&= e^{iS[q_{cl}]} C \prod_{n=1}^{\infty} \sqrt{\frac{\pi}{l_n}} = e^{iS[q_{cl}]} C \prod_{n=1}^{\infty} \sqrt{\frac{\pi}{\frac{-in^2\pi^2}{4T} \left(1 - \frac{\omega^2 T^2}{n^2\pi^2}\right)}}
\end{aligned} \tag{1.17}$$

We want to split the product into two parts to identify the factor in the given formula. In general $\prod_{n=1}^N a_n b_n = \left(\prod_{n=1}^N a_n \right) \left(\prod_{n=1}^N b_n \right)$ only holds for finite products and infinite products for which both factors converge separately. The first factor below doesn't converge. Thus we work with a finite N , then take the limit as $N \rightarrow \infty$

$$\left\langle q_b, T \middle| q_a, 0 \right\rangle = \lim_{N \rightarrow \infty} e^{iS[q_{cl}]} \left(C \prod_{n=1}^N \left(\frac{-4T}{in^2\pi} \right)^{\frac{1}{2}} \right) \sqrt{\frac{1}{\prod_{n=1}^N \left(1 - \frac{\omega^2 T^2}{n^2\pi^2} \right)}} \tag{1.18}$$

Define $C'_N = C \prod_{n=1}^N \left(\frac{-4T}{in^2\pi} \right)^{\frac{1}{2}}$ which is independent of q_a , q_b and ω , then we have

$$\left\langle q_b, T \middle| q_a, 0 \right\rangle = e^{iS[q_{cl}]} C'_N \sqrt{\frac{1}{\prod_{n=1}^N \left(1 - \frac{\omega^2 T^2}{n^2\pi^2} \right)}} \tag{1.19}$$

The ω factor converges and the first factor is divergent and independent of ω (it may be absorbed into normalisation). We are given

$$\frac{\sin z}{z} = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2\pi^2} \right).$$

where we can identify $z = \omega T$. Thus we have

$$\left\langle q_b, T \middle| q_a, 0 \right\rangle = e^{iS[q_{cl}]} C'_\infty \sqrt{\frac{\omega T}{\sin \omega T}} \tag{1.20}$$

Strictly speaking I have answered the what is explicitly stated in the question. The above answer should suffice and should be constitute the marked component of my answer.

However, for fun [please don't mark this :)], I calculated the precise form of C'_∞ then refer you to [A](#).¹

¹I believe there was a typo in the measure given in the question.

2 Question II

2.1 Part (a)

Given that $q = \{q_1, \dots, q_n\}$ and $j = \{j_1, \dots, j_n\}$. So $q^T \cdot j = q^T j = j^T \cdot q = j^T q$. we need to prove

$$-\frac{i}{2} q^T A q + i j \cdot q = -\frac{i}{2} (q - A^{-1} j)^T A (q - A^{-1} j) + \frac{i}{2} j^T A^{-1} j.$$

Equivalently we can divide by $-\frac{i}{2}$ and work with the left-hand side

$$\begin{aligned} (q - A^{-1} j)^T A (q - A^{-1} j) - j^T A^{-1} j &= q^T A q - 2j \cdot q \\ &= q^T A q - q^T A A^{-1} j - j^T (A^{-1})^T A q + j^T (A^{-1})^T A A^{-1} j - j^T A^{-1} j \\ &= q^T A q - q^T j - j^T A^{-1} A q + j^T A^{-1} j - j^T A^{-1} j \\ &= q^T A q - q^T j - j^T q \\ &= q^T A q - 2q \cdot j. \end{aligned}$$

where we have used the fact that $(AB)^T = B^T A^T$ and given that A is symmetric we have $A = A^T$ and $(A^{-1})^T = A^{-1}$.

We also used the fact that the transpose of a scalar is the same scalar.

2.2 Part (b)

We need to evaluate

$$I(j; n) = \int d^n q e^{(-\frac{i}{2} q^T A q + i j^T q)} \quad (2.1)$$

Define $v := q - A^{-1} j$

$$\begin{aligned} I(j; n) &= \int d^n v e^{-\frac{i}{2} v^T A v + \frac{i}{2} j^T A^{-1} j} \\ &= \int \prod_{m=1}^n dv_m e^{\frac{i}{2} j^T A^{-1} j} \exp \left[-\frac{i}{2} \sum_{k,l=1}^n v_k A_{kl} v_l \right] \\ &= \left(e^{\frac{i}{2} j^T A^{-1} j} \right) \int \prod_{m=1}^n dv_m \exp \left[-\frac{i}{2} \sum_{k,l=1}^n v_k A_{kl} v_l \right]. \end{aligned}$$

Since A is symmetric so is A^{-1} and thus there exists an orthogonal diagonalization of A : $A = S^T D S$ where $S S^T = S^T S = \mathbf{1}$ and D is a diagonal matrix whose entries are the eigenvalues of A i.e. $(\text{diag}(\lambda_1, \dots, \lambda_n))$. We have the form $v_k A_{kl} v_l \rightarrow v_k S_{kb}^T D_{bc} S_{cl} v_l$. Let $y = S v$, thus, $d^n y = d^n v$ and $v S^T D^{-1} S v = y^T D^{-1} y$

$$\begin{aligned} I(j; n) &= \left(e^{\frac{i}{2} j^T A^{-1} j} \right) \int \prod_{m=1}^n dy_m \exp \left[-\frac{i}{2} y^T D y \right] \\ &= \left(e^{\frac{i}{2} j^T A^{-1} j} \right) \int \prod_{m=1}^n dy_m \exp \left[-\frac{i}{2} \sum_{j=1}^n \lambda_j y_j^2 \right]. \end{aligned}$$

Using the standard Gaussian

$$\int_{-\infty}^{\infty} du e^{-au^2} = \sqrt{\frac{\pi}{a}} \quad (2.2)$$

$$\begin{aligned} I(j; n) &= \left(e^{\frac{i}{2} j^T A^{-1} j} \right) \int \prod_{m=1}^n dy_m \exp \left[-\frac{i}{2} \sum_{J=1}^n \lambda_J y_J^2 \right] \\ &= \left(e^{\frac{i}{2} j^T A^{-1} j} \right) \prod_{m=1}^n \left(\frac{2\pi}{i\lambda_J} \right)^{\frac{1}{2}} = \left(e^{\frac{i}{2} j^T A^{-1} j} \right) \frac{(-2\pi i)^{\frac{n}{2}}}{\sqrt{\det A}}. \end{aligned}$$

$$I(j; n) = e^{\frac{i}{2} j^T A^{-1} j} \frac{(-2\pi i)^{\frac{n}{2}}}{\sqrt{\det A}} \quad (2.3)$$

We notice if we set $j = 0$ we arrive at the given expression

$$I(0; n) = \frac{(-2\pi i)^{\frac{n}{2}}}{\sqrt{\det A}} \quad (2.4)$$

Futhermore, if we restrict $n = 1$ then A is just a scalar a

$$I(0; 1) = \sqrt{\frac{-2\pi i}{a}} = \sqrt{\frac{2\pi}{ia}} \quad (2.5)$$

which corresponds to a Gaussian with the extra prefactor $\frac{i}{2}$ as expected $\int dx e^{-\frac{i}{2} ax^2}$

3 Question III

3.1 Part (a)

An n point Green's function in a free scalar theory is given by

$$G^{(n)}(x_1, \dots, x_n) = \left\langle 0 \left| \hat{T} [\hat{\phi}(x_1), \dots, \hat{\phi}(x_n)] \right| 0 \right\rangle = \frac{1}{i^n} \frac{\delta}{\delta J(x_1)} \cdots \frac{\delta}{\delta J(x_n)} \frac{Z_{\text{free}}[J]}{Z_{\text{free}}[0]} \Big|_{J=0}.$$

We are interested in the case $n = 4$.

$$\begin{aligned} G^{(4)}(x_1, x_2, x_3, x_4) &= \left\langle 0 \left| \hat{T} [\hat{\phi}(x_1), \hat{\phi}(x_2), \hat{\phi}(x_3), \hat{\phi}(x_4)] \right| 0 \right\rangle \\ &= \left\{ \frac{1}{i^4} \frac{\delta}{\delta J(x_1)} \frac{\delta}{\delta J(x_2)} \frac{\delta}{\delta J(x_3)} \frac{\delta}{\delta J(x_4)} \frac{Z_{\text{free}}[J]}{Z_{\text{free}}[0]} \right\} \Big|_{J=0}. \end{aligned}$$

where in lectures we derived

$$Z_{\text{free}}[J] = Z_{\text{free}}[0] e^{-\frac{1}{2} \int d^2x d^2x' J(x') \Delta_F(x-x') J(x)}.$$

Thus, we can define

$$\begin{aligned} \frac{Z_{\text{free}}[J]}{Z_{\text{free}}[0]} &= e^{-\frac{1}{2} \int d^2x d^2x' J(x') \Delta_F(x-x') J(x)} := \exp \left[-\frac{1}{2} \Lambda[J] \right] \\ \delta_k &:= \frac{\delta}{\delta J(x_k)}. \end{aligned}$$

So we have

$$G^{(4)}(x_1, x_2, x_3, x_4) = \left\{ \prod_{k=1}^4 \delta_k \exp \left[-\frac{1}{2} \Lambda[J] \right] \right\} \Big|_{J=0}.$$

We notice that Λ is quadratic in J

$$\exp \left[-\frac{1}{2} \Lambda[J] \right] = 1 - \frac{1}{2} \Lambda + \frac{1}{8} \Lambda^2 + \dots \quad (3.1)$$

Since we take four functional derivatives terms less than $\mathcal{O}(\Lambda^2)$ vanish. Since we will set $J = 0$ the terms which have J remaining vanish i.e. terms of $\mathcal{O}(\Lambda^3)$. Thus the only remaining terms are quadratic in Λ .

$$\begin{aligned} G^{(4)}(x_1, x_2, x_3, x_4) &= \left\{ \frac{1}{8} \prod_{k=1}^4 \delta_k \Lambda^2[J] \right\} \Big|_{J=0} \\ \delta_k \Lambda &= 2 \int d^4x' \Delta_F(x_k - x') J(x') := 2g_k \\ \delta_l g_k &= \Delta_F(x_k - x_l) := \Delta_{kl}. \end{aligned}$$

where we used the fact $\frac{\delta J(x)}{\delta(x_l)} = \delta(x' - x_l)$ and $\delta_j \Delta_{kl} = 0$.
 Hopefully we can take the derivatives

$$\begin{aligned}\delta_1 \Lambda^2 &= 4\Lambda g_1 \\ \delta_2 [4\Lambda g_1] &= 8g_2 g_1 + 4\Lambda \Delta_{12} \\ \delta_3 [8g_2 g_1 + 4\Lambda \Delta_{12}] &= 8\Delta_{23} g_1 + 8g_2 \Delta_{13} + 8g_3 \Delta_{12} \\ \delta_4 [8\Delta_{23} g_1 + 8g_2 \Delta_{13} + 8g_3 \Delta_{12}] &= 8\Delta_{23} \Delta_{41} + 8\Delta_{13} \Delta_{24} + 8\Delta_{34} \Delta_{12}.\end{aligned}$$

Thus we arrive at

$$G^{(4)}(x_1, x_2, x_3, x_4) = \Delta_F(x_1 - x_2) \Delta_F(x_3 - x_4) + \Delta_F(x_1 - x_3) \Delta_F(x_2 - x_4) + \Delta_F(x_1 - x_4) \Delta_F(x_2 - x_3).$$

which agrees with the Feynman rules as shown below

$$\begin{aligned}G_{\text{free}}^{(4)}(x_1, x_2, x_3, x_4) &= \begin{array}{c} 1 \\ \bullet \\ | \\ \bullet \\ 2 \end{array} \begin{array}{c} 3 \\ \bullet \\ | \\ \bullet \\ 4 \end{array} + \begin{array}{c} 1 \quad 3 \\ \bullet \quad \bullet \\ \hline \bullet \quad \bullet \\ 2 \quad 4 \end{array} + \begin{array}{c} 1 \quad 3 \\ \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \quad \bullet \\ 2 \quad 4 \end{array} \\ &= \Delta_F(x_1 - x_2) \Delta_F(x_3 - x_4) + \Delta_F(x_1 - x_3) \Delta_F(x_2 - x_4) + \Delta_F(x_1 - x_4) \Delta_F(x_2 - x_3).\end{aligned}$$

3.2 Part (b)

We want to find a 6 point greens function for a free scalar field. Since we know for an $2n$ point function there exists $\left(\frac{(2n)!}{2^n n!}\right)$ terms each of which has a corresponding Feynman diagram. A 6

point function thus has $\left(\frac{6!}{2^3 3!}\right) = 15$ terms. The answer is given in terms of $\Delta_F(x_i - x_j) := \Delta_{ij}$.

$$\begin{aligned}
G_{\text{free}}^{(6)}(x_1, \dots, x_6) = & \begin{array}{c} 1 \quad 3 \quad 5 \\ \bullet \quad \bullet \quad \bullet \\ | \quad | \quad | \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} + \begin{array}{c} 1 \quad 3 \quad 5 \\ \bullet \quad \bullet \quad \bullet \\ | \quad \text{---} \quad | \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} + \begin{array}{c} 1 \quad 3 \quad 5 \\ \bullet \quad \bullet \quad \bullet \\ | \quad \diagup \quad \diagdown \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} \\
& + \begin{array}{c} 1 \quad 3 \quad 5 \\ \text{---} \quad \bullet \quad | \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} + \begin{array}{c} 1 \quad 3 \quad 5 \\ \text{---} \quad \bullet \quad \diagup \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} + \begin{array}{c} 1 \quad 3 \quad 5 \\ \text{---} \quad \bullet \quad \diagup \quad \diagdown \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} \\
& + \begin{array}{c} 1 \quad 3 \quad 5 \\ \diagdown \quad \bullet \quad | \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} + \begin{array}{c} 1 \quad 3 \quad 5 \\ \diagdown \quad \bullet \quad \diagup \quad \diagdown \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} + \begin{array}{c} 1 \quad 3 \quad 5 \\ \diagdown \quad \bullet \quad \text{---} \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} \\
& + \begin{array}{c} 1 \quad 3 \quad 5 \\ \text{---} \quad \bullet \quad \text{---} \\ \diagup \quad \bullet \quad \bullet \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} + \begin{array}{c} 1 \quad 3 \quad 5 \\ \text{---} \quad \bullet \quad \diagdown \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} + \begin{array}{c} 1 \quad 3 \quad 5 \\ \text{---} \quad \bullet \quad | \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} \\
& + \begin{array}{c} 1 \quad 3 \quad 5 \\ \diagdown \quad \bullet \quad \diagup \\ \diagdown \quad \bullet \quad \diagdown \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} + \begin{array}{c} 1 \quad 3 \quad 5 \\ \diagdown \quad \bullet \quad \text{---} \\ \text{---} \quad \bullet \quad \bullet \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} + \begin{array}{c} 1 \quad 3 \quad 5 \\ \diagdown \quad \bullet \quad \diagdown \quad \diagup \\ \bullet \quad \bullet \quad \bullet \\ 2 \quad 4 \quad 6 \end{array} \\
= & \Delta_{12}\Delta_{34}\Delta_{56} + \Delta_{12}\Delta_{35}\Delta_{46} + \Delta_{12}\Delta_{36}\Delta_{45} \\
& + \Delta_{13}\Delta_{24}\Delta_{56} + \Delta_{13}\Delta_{25}\Delta_{46} + \Delta_{13}\Delta_{26}\Delta_{45} \\
& + \Delta_{14}\Delta_{23}\Delta_{56} + \Delta_{14}\Delta_{25}\Delta_{36} + \Delta_{14}\Delta_{26}\Delta_{35} \\
& + \Delta_{15}\Delta_{23}\Delta_{46} + \Delta_{15}\Delta_{24}\Delta_{36} + \Delta_{15}\Delta_{26}\Delta_{34} \\
& + \Delta_{16}\Delta_{23}\Delta_{45} + \Delta_{16}\Delta_{24}\Delta_{35} + \Delta_{16}\Delta_{25}\Delta_{34}.
\end{aligned}$$

A Appendix

The following is for fun and not part of the assignment.

The result should match that of Question I Part (e) should match Question I Part (c) and since C'_∞ is independent of ω we can solve for it.

In the limit $\omega \rightarrow 0$ we have $\hat{H} = \frac{\hat{p}^2}{2}$ (a free particle), where $\dot{q} = p$.

$$\begin{aligned} \left\langle q_b \left| e^{-i\hat{H}t} \right| q_a \right\rangle &= \int \frac{dp}{2\pi} e^{ip(q_b - q_a)} e^{-i\frac{p^2}{2}T} \\ &= \int \frac{dp}{2\pi} e^{-\frac{iT}{2} \left(p + \frac{q_b - q_a}{T} \right)^2} e^{\frac{i(q_b - q_a)^2}{2T}} \\ &= e^{\frac{i(q_b - q_a)^2}{2T}} \sqrt{\frac{1}{2\pi iT}}. \end{aligned}$$

Now if we look at the answer from Part (e)

$$\left\langle q_b, T \left| q_a, 0 \right\rangle = e^{iS[q_{cl}]} C'_\infty \sqrt{\frac{\omega T}{\sin \omega T}} \quad (\text{A.1})$$

if we take $\omega \rightarrow 0$ we should match the above result. We know from Part (c) that for $\omega \rightarrow 0$, $S(q_{cl}) = \frac{1}{2T} (q_a - q_b)^2 + \mathcal{O}(\omega^2)$. Equating these gives

$$e^{\frac{i(q_b - q_a)^2}{2T}} \sqrt{\frac{1}{2\pi iT}} = e^{i\frac{1}{2T}(q_a - q_b)^2} C'_\infty \sqrt{\frac{\omega T}{\sin \omega T}} \quad (\text{A.2})$$

Thus since for $\omega \rightarrow 0$, $\sqrt{\frac{\omega T}{\sin \omega T}} \rightarrow 1$ we have

$$C'_\infty = \sqrt{\frac{1}{2\pi iT}} \quad (\text{A.3})$$